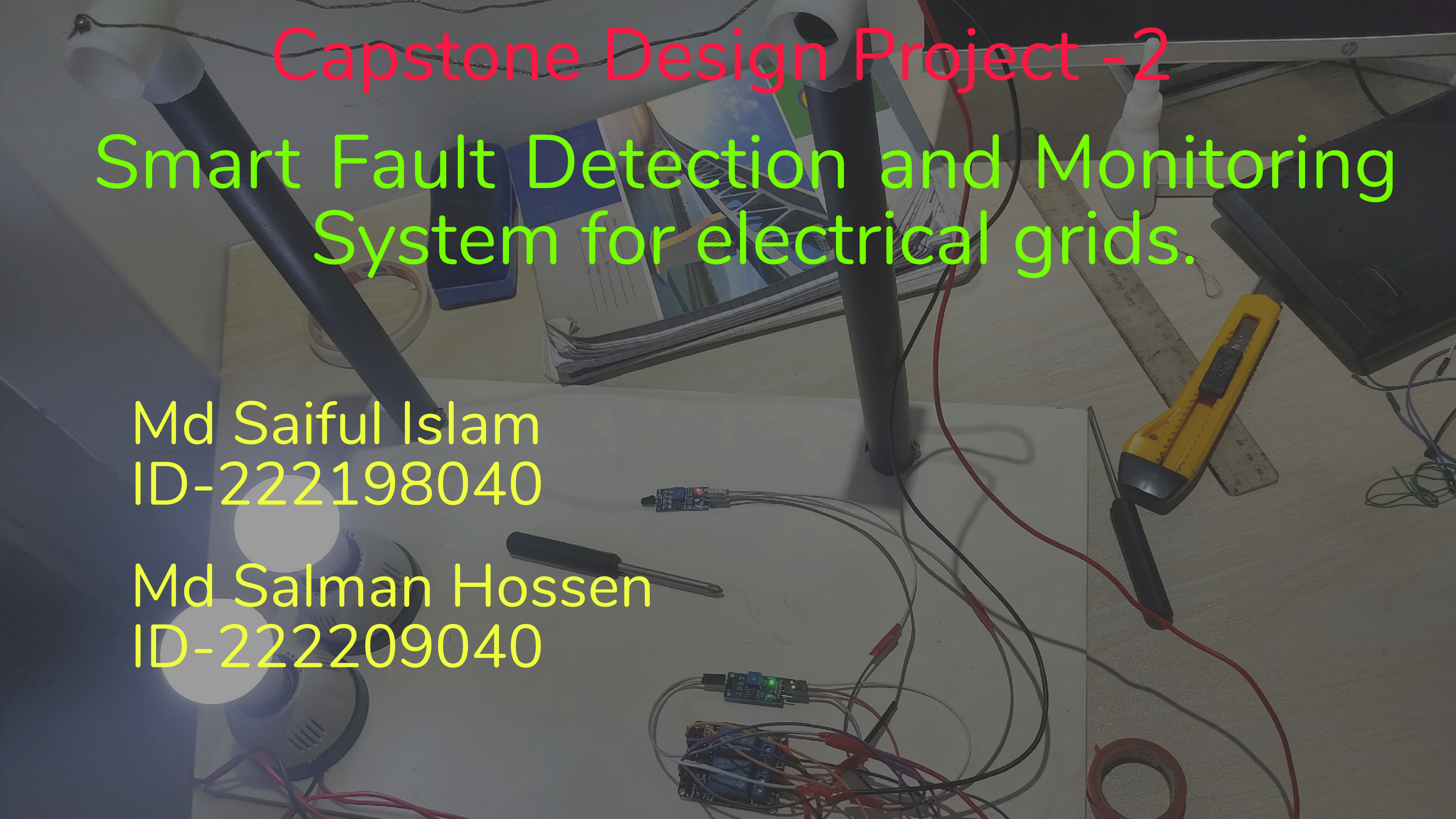


Capstone Design Project -2

Smart Fault Detection and Monitoring System for electrical grids.

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Introduction :

In modern power distribution, ensuring the reliability of "Grid Lines" is critical. Traditional systems often rely on manual inspection or basic circuit breakers that only react after a catastrophic short circuit has occurred. This project introduces an automated monitoring and protection system designed to detect environmental hazards—specifically fire and moisture—that precede or cause electrical line faults.

Components :

1. Double channel relay (5v)
2. copper wire
3. Moisture sensor
4. Flame sensor
5. Charging module
6. Bulb (5watt)
7. Switch (DC)

Components in Brief :

the 5V Relay Module acts as the "bridge" between the low-power brain (your sensors/battery) and the high-power load (the light bulbs).



Think of it as a digitally controlled switch. Here are the three main reasons why it is essential for your line fault detection system:

1. Electrical Isolation (Safety)
2. Signal Amplification
3. Automated Logic (The "Protective" Action)

Flame sensor:

It detects "Arcing Faults" or fires near the grid lines that could melt the wires.



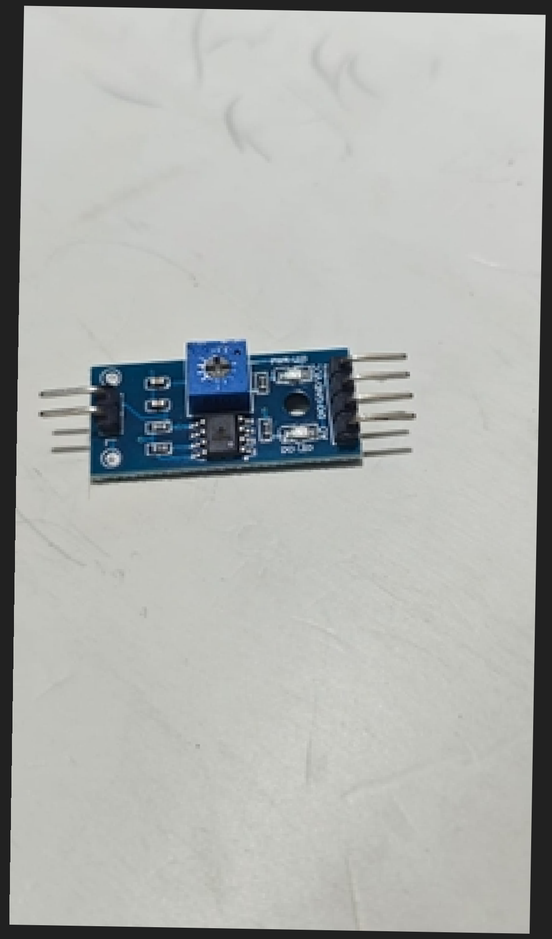
This sensor uses a specialized phototransistor that is highly sensitive to Infrared (IR) light. Flames emit a specific wavelength of IR light (usually between 760nm and 1100nm).

When the sensor "sees" this specific light frequency, its resistance drops. This allows a small signal to pass to your control circuit, which then triggers the relay to cut the power

Moisture sensor :

It prevents "Leakage Faults" caused by rain, flooding, or condensation inside electrical panels.

Water bridges the gap. Because water is a conductor, electricity flows across the legs. The sensor detects this "leakage" and tells the relay: "Fault detected! Open the switch!"



Working Principle :

The project is a smart safety switch. It constantly monitors the environment around power lines and automatically cuts the electricity if it detects a dangerous condition.

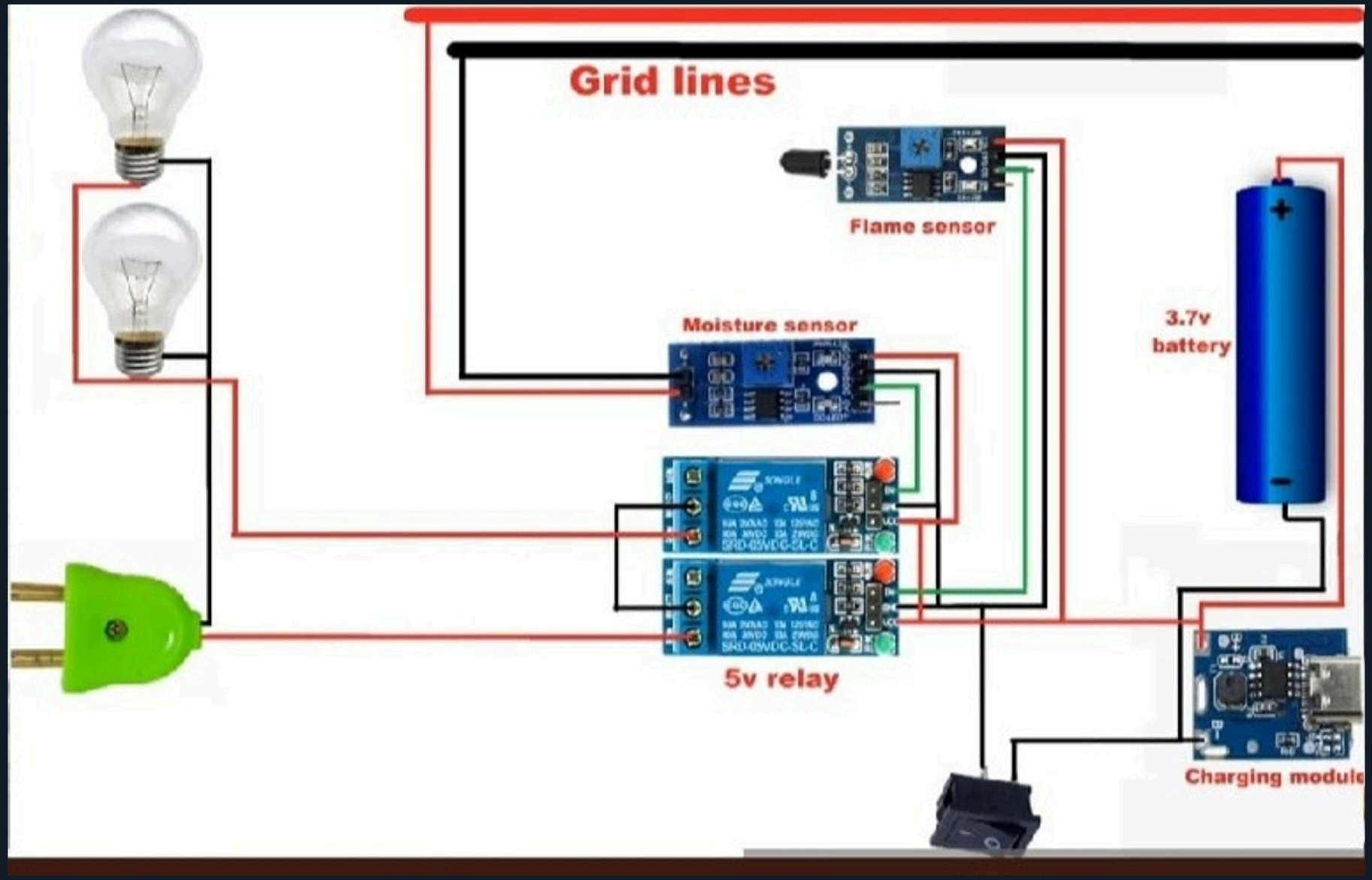
The 3-Step Process

Sensing: The Flame Sensor watches for fire/sparks, and the Moisture Sensor watches for water/leaks.

Processing: If either sensor detects a threat, it sends a "Danger" signal to the Relay.

Action: The Relay acts as a mechanical switch. It instantly "opens" the circuit, turning off the light bulbs (the load) to prevent a fire or short circuit.

Circuit Diagram :



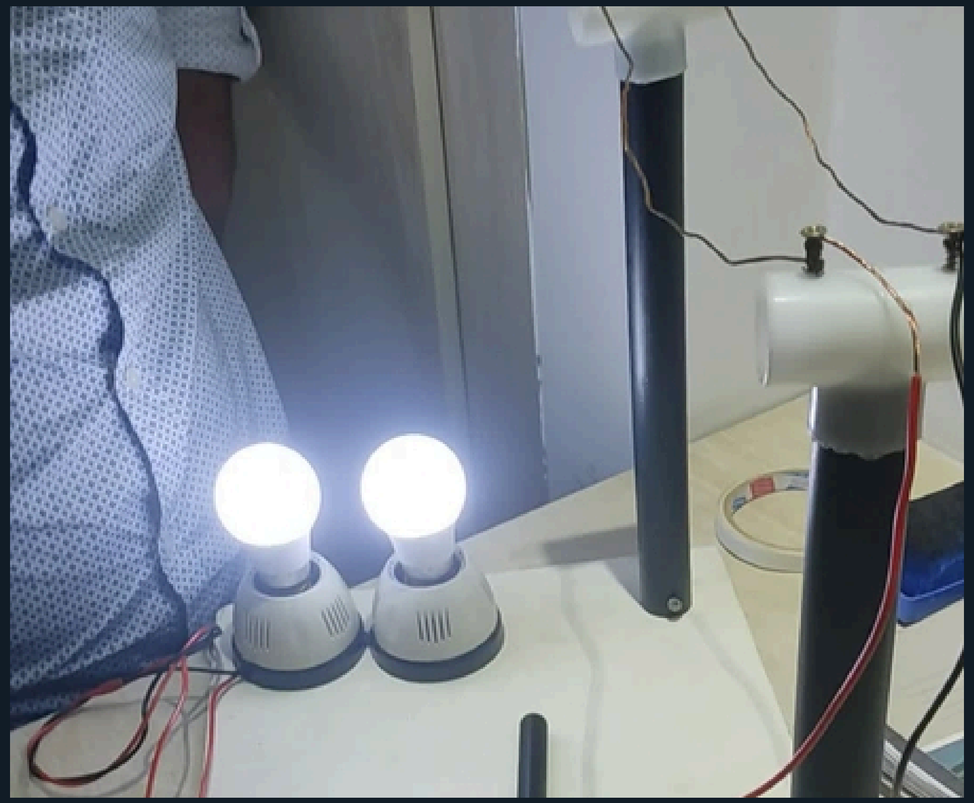
Significance :

The significance of this project can be summarized in three short points:

Proactive Protection: Unlike traditional fuses that only blow after a short circuit occurs, this system detects the cause (fire or water) and cuts power before the damage happens.

Reduced Fire Hazards: By instantly isolating the electrical line when a flame or spark is detected, it prevents small electrical faults from turning into large-scale building or forest fires

Cost Efficiency: It protects expensive appliances and grid infrastructure from water damage and power surges, reducing the need for costly repairs and manual inspections.



Thank you